MILITARY STANDARD

MIL-STD-1320-212 (NAVY)

2 JULY 1984

TRUCKLOADING ROCKET MOTOR FOR MISSILE, AGM-88A (HARM) IN CNU-354/E CONTAINER

CONTAINER DATA DODIC/NALC

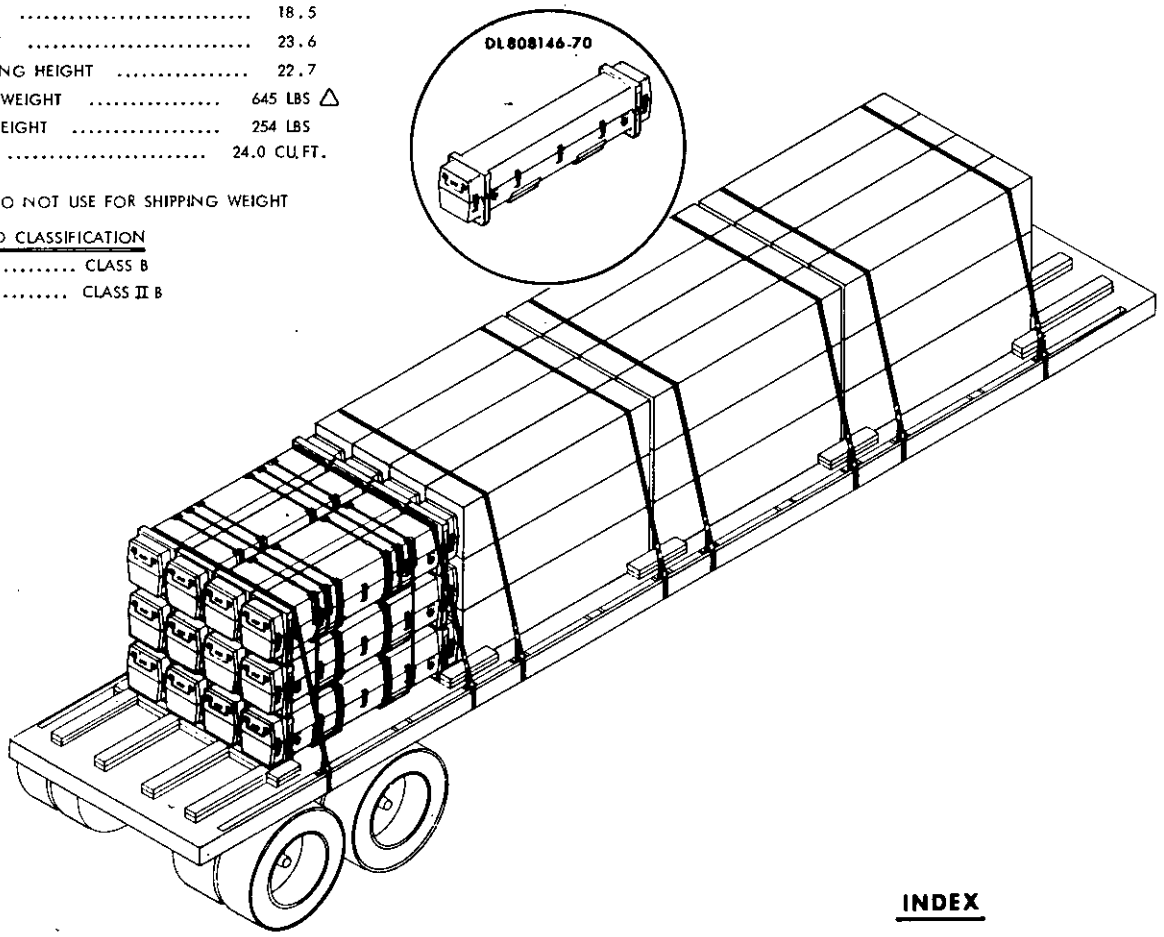
NUMBER OF MOTORS PER CONTAINER	1
LENGTH	95.0
WIDTH	18.5
HEIGHT	23.6
STACKING HEIGHT	22.7
GROSS WEIGHT	645 LBS △
TARE WEIGHT	254 LBS
CUBE	24.0 CU.FT.

△ DO NOT USE FOR SHIPPING WEIGHT

HAZARD CLASSIFICATION

DOT ..... CLASS B

C.G. ..... CLASS II B



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL 40-45 FT TRAILER	2,3&4
LTL 40-45 FT TRAILER	3,5&6
UNITIZING	7

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

W S Gray NWHC 6/27/84
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J E Kelly 6/29/84
SIGNATURE AIR SYSCOM BY DIRECTION DATE

ORIGINATOR Charles J McBride 6/28/84 DATE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THE DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE AGM-88A ROCKET MOTOR IN CONTAINER, SHIPPING AND STORAGE CNU-354/E.

2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 TO 45 FT FLATBED TRAILERS. THE TRAILER AXLES SHALL BE LOCATED IN THE WESTERN POSITION (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.

3. THE CONTAINERS CAN BE UNITIZED AS MULTIPLES OF SIX (6), FOUR (4), AND TWO (2) AS SHOWN ON PAGE 7.

4. ONLY 2 X .050 GALVANIZED STEEL STRAPPING SHALL BE USED FOR TIEDOWNS. CHAINS ARE NOT AUTHORIZED. STRAPPING SHALL MEET THE REQUIREMENTS OF THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 AND BE APPLIED AS SHOWN IN THIS DOCUMENT.

5. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.

6. AFTER BLOCKING, BRACING AND TIEDOWNS HAVE BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVES A" PLACARDS TO BOTH SIDES, FRONT AND BACK OF TRAILER.

7. UNLESS OTHERWISE SPECIFIED, MATERIALS SHALL BE IN ACCORDANCE WITH THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320.

8. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320

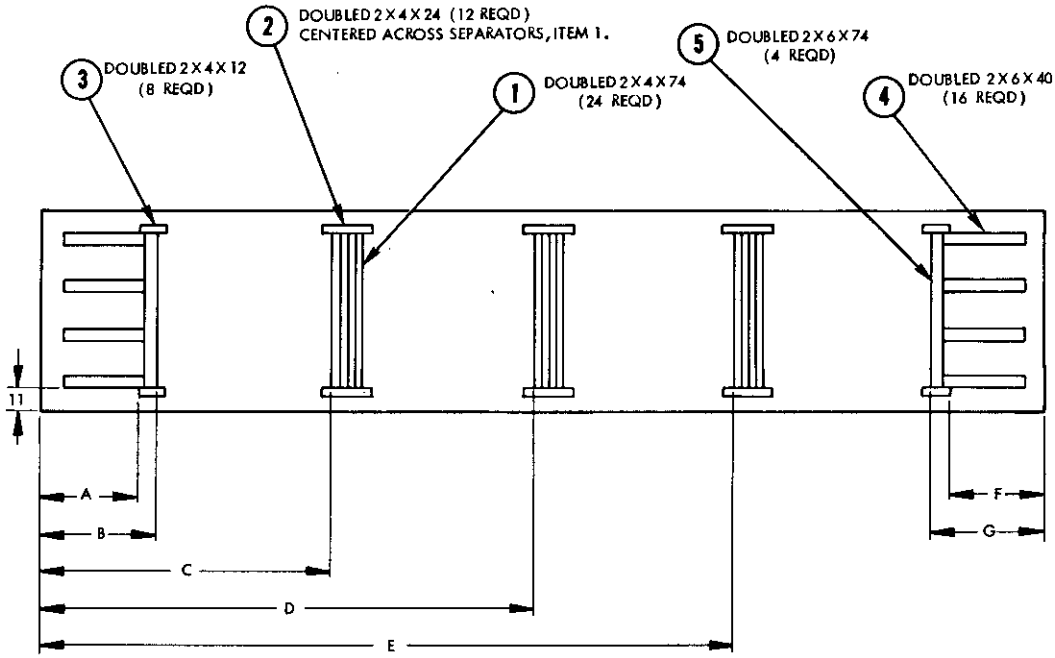
FTL 40,42,44 AND 45 FT TRAILER(FLATBED)

1. A FULL TRUCKLOAD CONSISTS OF 48 CNU-354/E CONTAINERS.

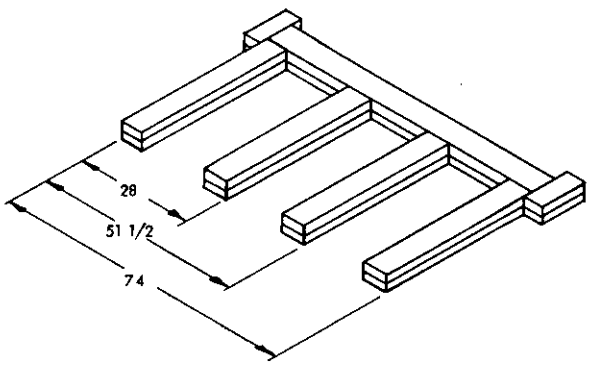
2. ADJACENT UNIT LOADS OF EITHER FOUR OR SIX CONTAINERS SHOULD BE STRAPPED TOGETHER WITH TWO 1 1/4 X .035 STEEL STRAPS WHICH ARE THREADED THROUGH THE UPPERMOST CONTAINER FORK LIFT POCKETS AND OVER THE TOP COVERS (4 WIDE), TENSIONED AND DOUBLE NOTCH SEALED.

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PREPOSITION BLOCKING



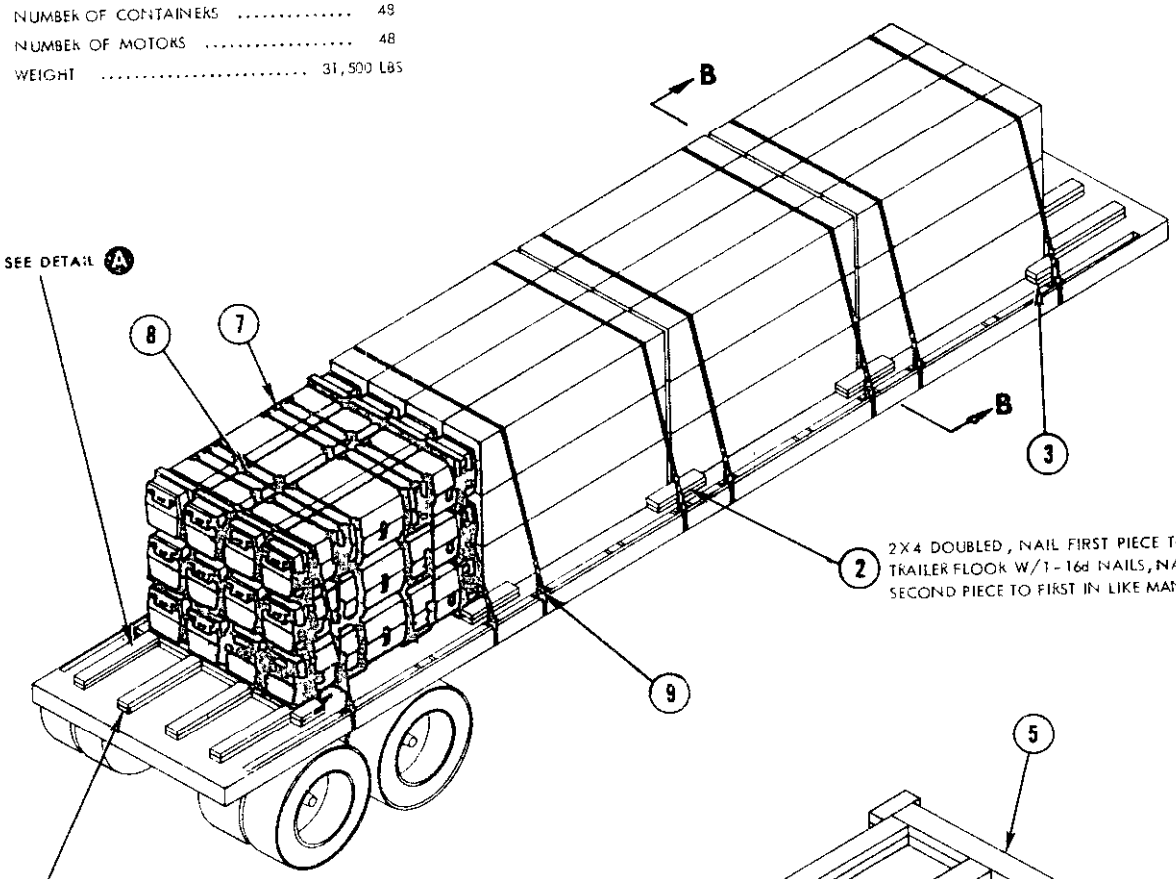
TRAILER LENGTH	A	B	C	D	E	F	G
40 FT	47	55 1/2	137 1/2	233 1/2	329 1/2	46	54 1/2
42 FT	59	67 1/2	149 1/2	245 1/2	341 1/2	58	66 1/2
44 FT	71	79 1/2	161 1/2	257 1/2	353 1/2	70	78 1/2
45 FT	77	85 1/2	167 1/2	263 1/2	359 1/2	76	84 1/2

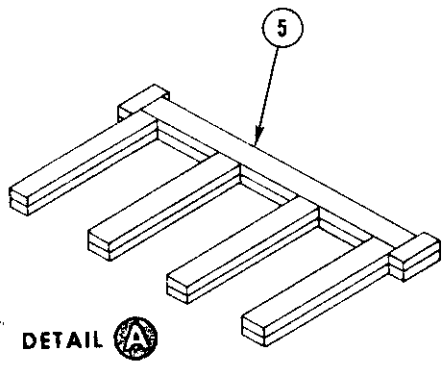


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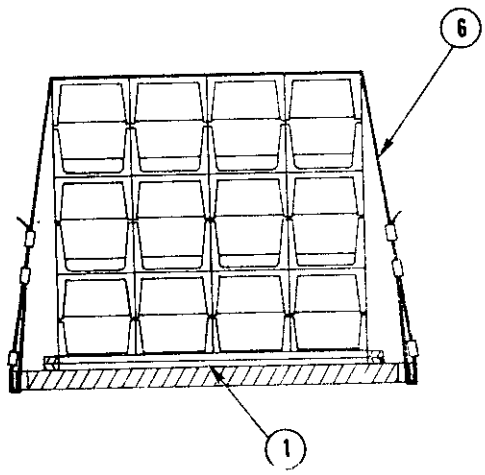
TRUCKLOAD DATA

NUMBER OF CONTAINERS 48
NUMBER OF MOTORS 48
WEIGHT ..... 31,500 LBS





DETAIL A



SECTION B-B

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
48	9	SEAL	SEE NOTE 4	2
8	8	SEAL	SEE NOTE 4	1 1/4
8	7	STACKING STRAP	SEE NOTE 4 & 8	1 1/4 X .035 X 17 FT
8	6	TIE DOWN STRAP	SEE NOTE 4 & 8	2 X .050 X 29 FT
4	5	CROSSMEMBER	SEE NOTE 8 AND FIELD NOTES	2 X 6 X 74
16	4	BACKUP CLEAT	SEE NOTE 8 AND FIELD NOTES	2 X 6 X 40
8	3	SLEEPER, END	SEE NOTE 8 AND FIELD NOTES	2 X 4 X 12
12	2	SLEEPER, SIDE	SEE NOTE 8 AND FIELD NOTES	2 X 4 X 24
24	1	CROSSMEMBER	SEE NOTE 8 AND FIELD NOTES	2 X 4 X 74

LIST OF MATERIALS

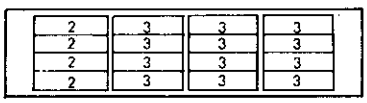
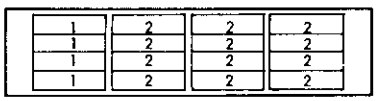
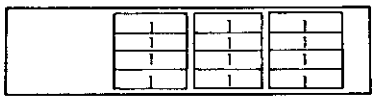
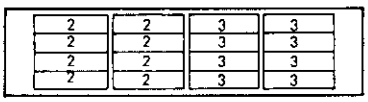
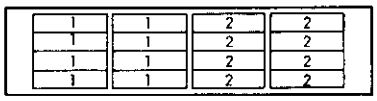
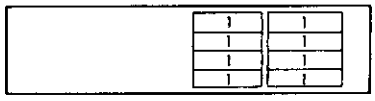
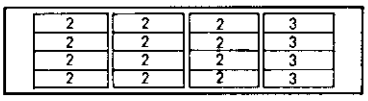
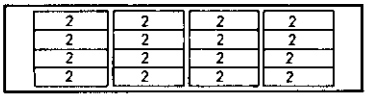
FTL 40 FT TRAILER (FLATBED)

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LTL 40-45 FT TRAILERS (FLATBED)

1. FOR 16 OR MORE CONTAINERS PER TRUCK LOAD, USE PRE-POSITIONED BLOCKING AS SHOWN ON PAGE 3, ADJUST DIMENSIONS AS REQUIRED FOR FEWER THAN 16 CONTAINERS
2. ODD NUMBERS OF CONTAINERS (15 OR LESS ONLY) MAY ALSO BE SHIPPED BY ALTERING PRE POSITIONED BLOCKING.
3. WHEN REQUIRED TO SHIP LESS THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF CONTAINERS TO BE SHIPPED FROM TABLE I.
4. POSITION CONTAINERS ON TRAILER AS SHOWN IN TABLE. BLOCK AND TIEDOWN LADING USING THE PRINCIPLES SHOWN ON PAGE 3 AND THE GENERAL DOCUMENT, MIL-STD-1320.
5. THE LTL PLAN SHOWN ON PAGE 6 IS THE CORRECT LESS-THAN-TRUCKLOAD FOR THE QUANTITY OF CONTAINERS SHOWN. THIS DOES NOT LIMIT SHIPMENTS TO THIS QUANTITY, SINCE LTL SHIPMENTS MAY CONSIST OF A GRATER OR LESSER NUMBER OF CONTAINERS.

TABLE I

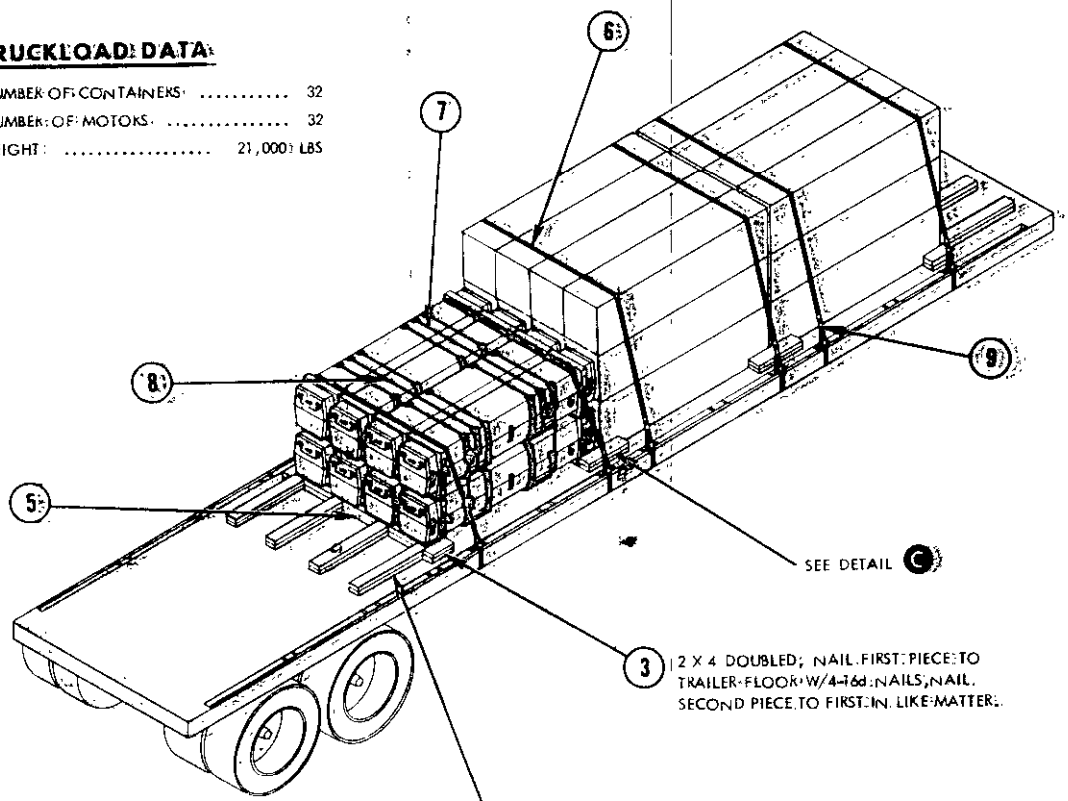
NO. OF CONTAINERS	LOAD PATTERN △ FWD →	NO. OF CONTAINERS	LOAD PATTERN △ FWD →	NO. OF CONTAINERS	LOAD PATTERN △ FWD →
44		28		12	
40		24		8	
36		20		4	
32		16			

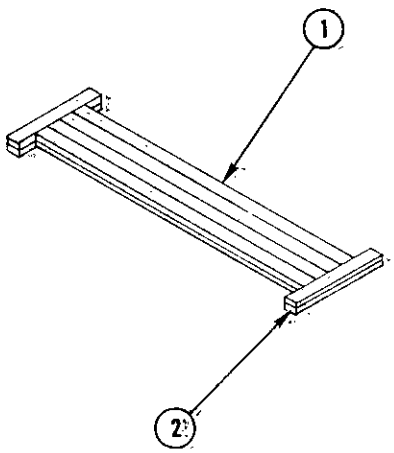
△ NUMBER IN LOAD PATTERN BLOCK INDICATES NUMBER OF CONTAINERS IN A LAYER.

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TRUCKLOAD DATA

NUMBER OF CONTAINERS 32
NUMBER OF MOTORS 32
WEIGHT ..... 21,000 LBS





DETAIL C

REQ'D	ITEM	DESCRIPTION	MAT'L/D.W.G.	DIMENSIONS
36	9	SEAL	SEE G.NOTE 4	2"
6	8	SEAL	SEE G.NOTE 4	1 1/4"
6	7	STACKING STRAP	SEE G.NOTE 4 & 8	1 1/4" X .035 X 17 FT
6	6	TIE DOWN STRAP	SEE G.NOTE 4 & 8	2" X .050 X .29 FT *
4	5	CROSSMEMBER	SEE G. NOTE 8 AND FIELD NOTES	2" X 6" X 74"
16	4	BACKUP CLEAT	SEE G. NOTE 8 AND FIELD NOTES	2" X 6" X 40"
8	3	SLEEPER END	SEE G. NOTE 8 AND FIELD NOTES	2" X 4" X 12"
8	2	SLEEPER, SIDE	SEE G. NOTE 8 AND FIELD NOTES	2" X 4" X 24"
16	1	CROSSMEMBER	SEE G. NOTE 8 AND FIELD NOTES	2" X 4" X 74"

LIST OF MATERIALS

G.NOTE = GENERAL NOTES * REAR STACK TIE DOWN = 25 FT

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UNITIZING PROCEDURE

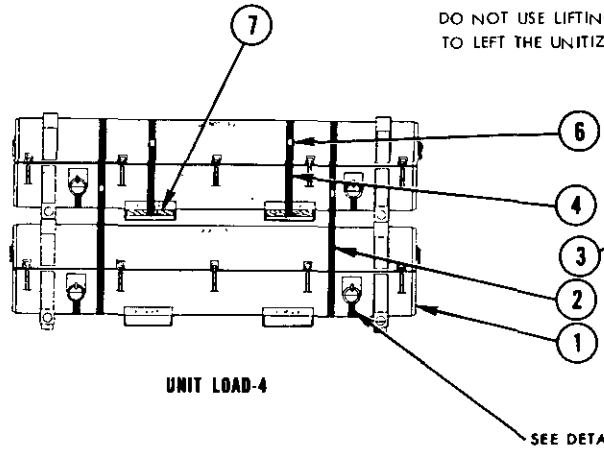
1. LOOP STRAP, ITEM 5, THROUGH LIFTING EYES OF EACH CONTAINER, ITEM 1, SO THAT EYES ARE IN DOWN POSITION. TENSION AND DOUBLE NOTCH SEAL, ITEM 6 .
2. STACK TWO CONTAINERS, THEN PLACE TWO STACKS SIDE BY SIDE, SO THAT MOISTURE INDICATORS ARE ON SAME ENDS.
3. THREAD STRAP, ITEM 4, THROUGH FORK POCKETS.
4. PLACE 2 X 8 X 31 BATTEN, ITEM 7, ON TOP OF STRAPPING, ITEM 4 .
5. PUT STRAP, ITEM 4, AROUND TOP LAYER OF CONTAINERS, TENSION AND DOUBLE NOTCH SEAL .
6. PUT UNIT LOAD STRAP, ITEM 2, AROUND UNIT LOAD, TENSION AND DOUBLE NOTCH SEAL .

NOTE: STEPS 1 THROUGH 6 DESCRIBE UNITIZATION OF FOUR (4) CONTAINERS. TO UNITIZE SIX (6) CONTAINERS PROCEED WITH STEPS 1 THROUGH 6 AND THEN FOLLOW STEPS 7 AND 8 .

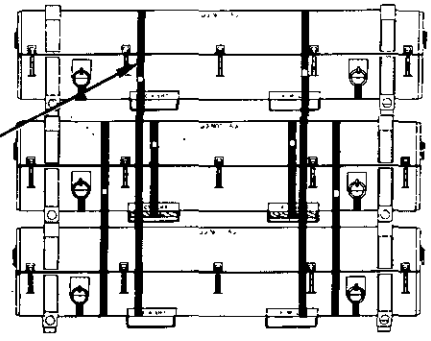
7. PLACE TWO MORE CONTAINERS, ONE ON EACH STACK, MAKING SURE STACKING FEATURES ARE SECURELY ENGAGED .
8. THREAD UNIT LOAD STRAPPING, ITEM 3, THROUGH THE BOTTOM CONTAINER FORK LIFT POCKETS, OVER THE TOP CONTAINERS (THREE HIGH), TENSION AND DOUBLE NOTCH SEAL .

CAUTION

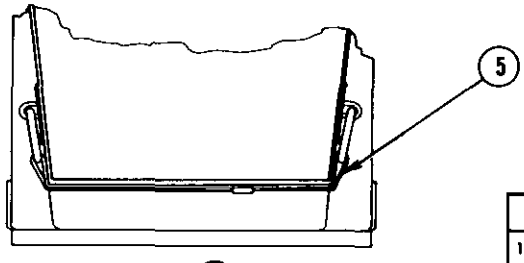
DO NOT USE LIFTING EYES TO LEFT THE UNITIZED LOAD.



UNIT LOAD-4



UNIT LOAD-6



DETAIL A

ROTATED 90°

DE-UNITIZING PROCEDURE

1. CUT AND REMOVE STRAPPING.
2. REMOVE BATTENS
3. REMOVE CONTAINERS

NOTE

1. ALL MATERIALS SHALL BE AS SPECIFIED IN MIL-STD-1323.

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
2	7	BATTEN	SEE NOTE 1	2 X 8 X 31
12 OR 16	6	SEAL	SEE NOTE 1	3/4
8 OR 12	5	STRAPPING (LOOP FOR LIFT EYES)	SEE NOTE 1	3/4 X .035 X 4 FT 6
2	4	STRAPPING (BATTEN RETAINING)	SEE NOTE 1	3/4 X .035 X 10 FT
2	3	STRAPPING (UNIT LOAD -6)	SEE NOTE 1	3/4 X .035 X 17 FT 6
2	2	STRAPPING (UNIT LOAD-4)	SEE NOTE 1	3/4 X .035 X 13 FT 6
4 OR 6	1	CONTAINER	18894-808146	95 X 18 1/2 X 23 5/8

LIST OF MATERIALS

REVIEW ACTIVITY:
NAVY-OS, AS

PREPAIRING ACTIVITY:
NAVY-OS
(PROJECT NO. 8140-N584)

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